Documentation Tracker · Repository & Mapping Model

Q: What does the tracker know about the repository at one commit, and which authored links decide what goes to human review?
Analysis exists only once both snapshots load; the saved run file (M4) keeps commits, status, findings and errors, not snapshots. Keys are paths and names.

RUN · IN MEMORY

Analysis

Exists only once both snapshots load; failures go to M4.

base_snapshot: `Snapshot | null`
Model at base; null for a first audit.

head_snapshot: `Snapshot`
Model at head.

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Snapshot

Repository model at one commit.

commit: `string`
Resolved Git commit identifier.

sections: `DocSection[]`
Markdown sections, plus one whole-file section each.

dependencies: `Dependency[]`
Code-to-code edges found at this commit.

root: `Folder`
Top folder (path ""); where code selectors start.

unparsed: `string[]`
Code files that failed to parse; reported as skipped.

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DocSection

One Markdown heading and everything under it.

doc_path: `string`
Path of the Markdown file.

heading_path: `string[]`
Headings from the top down; [] is the whole-file section.

start_line: `int`
Heading line; 1 for the whole-file section.

end_line: `int`
Line before the next equal-or-higher heading, or EOF.

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Dependency

A call or inheritance edge; name-based in v1.

source: `string`
Selector of the code that depends (caller, subclass).

target: `string`
Selector of what it uses (callee, base type).

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Folder

A directory in the repository.

name: `string`
Last path segment; display only.

path: `string`
Repo-relative, '/'-separated; the root is "".

files: `File[]`
Files directly inside this folder.

folders: `Folder[]`
Subfolders directly inside this folder.

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File

Lexical tree: each symbol sits in the file holding its text.

path: `string`
Repo-relative path; the file's identity in a revision.

functions: `Function[]`
Top-level callables, incl. methods declared outside a type.

classes: `Class[]`
Top-level type containers declared in this file.

AUTHORED · MAPPING FILE IN THE REPO

DocLink

One entry in the mapping file; read at base and head.

doc_path: `string`
Key part 1: the section's Markdown file.

heading_path: `string[]`
Key part 2: the section's heading path.

code: `string[]`
Code the section describes, as selectors below.

CODE SELECTOR — one string per target

`src/model/`	folder (trailing /)
`src/model/scoring.py`	whole file
`src/model/scoring.py::score`	top-level function or class
`src/model/scoring.py::Scorer.score`	method of a type

RESOLUTION RULES

Each revision is resolved on its own; renames are not followed.
A selector that matches overloads covers all of them.
Nested declarations count as part of their parent's span.
A heading path repeated within one file is an invalid mapping.

One section → many targets; one target → many links.

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Function

Any named callable: function, method, constructor.

name: `string`
Local name; a method outside its type body is Type.method.

start_line: `int`
First line, including decorators or annotations.

end_line: `int`
Last line of the body (inclusive).

signature: `string`
Declaration header text, stored verbatim.

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Class

Any named type: class, struct, interface, trait, enum.

name: `string`
Local name; the key inside its file.

start_line: `int`
First line, including decorators or annotations.

end_line: `int`
Last line of the body, including its methods.

methods: `Function[]`
Callables declared inside this type's body.

RULE · WHICH LINKS GO TO REVIEW (M3, SYMBOL LEVEL)

Base → head candidates

`git diff -U0 base head → changed line ranges`
code changed: a target's span overlaps a hunk on either side,
it exists at base only, or a target it depends on changed (1 hop).
doc changed: the section's span overlaps a hunk.

no base (first audit)	`resolution checks only`
link kept, section gone, target kept	`MISSING_DOCUMENTATION`
section and targets gone, or link removed	`no finding`
target or heading does not resolve	`SKIPPED · INVALID_MAPPING`
code file failed to parse	`SKIPPED · PARSE_FAILED`
code changed, doc not	`OUTDATED_DOCUMENTATION`
both changed, or only doc changed	`not flagged in v1 (gap)`
changed code in no link	`listed in changes[] only`

Every finding is a candidate for human review (PENDING_REVIEW).
Hunk overlap, not line numbers: edits above a function only shift it.

→ Contains an object or list

⇢ Resolved by key: path · heading_path · path::name (also Dependency source and target)

string = text · int = whole number · [] = list · lines are 1-based and inclusive · a parent's span includes its children (as LSP document symbols do)
Function and Class come from per-language tree-sitter adapters. Methods declared outside their type (Go receivers, Rust impl, C++) stay in their own file as Type.method. v1 parses Python only.